The Algorithm

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The FSRS (Free Spaced Repetition Scheduler) algorithm originates in the DHP model from MaiMemo, which is a variant of the DSR (Difficulty, Stability, Retrievability) model used to predict memory states.

Here is a visualizer to preview the interval with specific parameters and review history: Anki FSRS Visualizer (open-spaced-repetition.github.io).

In case you find this article difficult to understand, perhaps you will like this more: https://expertium.github.io/Algorithm.html.

Symbol

- R: Retrievability (probability of recall)
- S: Stability (interval when $R = 90$)
  - $S_r^{'}$: new stability after recall
  - $S_f^{'}$: new stability after forgetting
- D: Difficulty ($D \in [1, 10]$)
- G: Grade (rating at Anki):
  - 1: `again`
  - 2: `hard`
  - 3: `good`
  - 4: `easy`

FSRS-6

Default parameters

```
[0.212, 1.2931, 2.3065, 8.2956, 6.4133, 0.8334, 3.0194, 0.001, 1.8722, 0.1666, 0
```

Formula

> The w_i denotes w[i]. This version uses 21 parameters. The memory state is represented by Stability (S) and Difficulty (D).

The formula of stability after a same-day review is changed in this update:

$$S^{'}(S,G) = S \cdot e^{w_{17} \cdot (G - 3 + w_{18})} \cdot S^{-w_{19}}.$$

So the S increases faster when it's small and slower when it's large. S will converge when $SInc = e^{w_{17} \cdot (G - 3 + w_{18})} \cdot S^{-w_{19}} = 1$.

In practice, we should ensure that $SInc \geq 1$ when $G \geq 3$.

The forgetting curve's decay is trainable:

$$R(t,S) = \left(1 + factor \cdot \frac{t}{S}\right)^{-w_{20}},$$

where $factor = 0.9^{-\frac{1}{w_{20}}} - 1$ to ensure $R(S,S) = 90\%$.

FSRS-5

Default parameters

```
[0.40255, 1.18385, 3.173, 15.69105, 7.1949, 0.5345, 1.4604, 0.0046, 1.54575, 0.1
```

Formula

> The w_i denotes w[i]. This version uses 19 parameters. The memory state is represented by Stability (S) and Difficulty (D).

The stability after a same-day review:

$$S^{'}(S,G) = S \cdot e^{w_{17} \cdot (G - 3 + w_{18})}$$

The initial difficulty after the first rating:

$$D_0(G) = w_4 - e^{w_5 \cdot (G-1)} + 1,$$

where $w_4 = D_0(1)$, i.e., the initial difficulty when the first rating is `Again` .

Linear Damping for the new difficulty after review:

$$\Delta D(G) = -w_6 \cdot (G-3)$$

$$D' = D + \Delta D \cdot \frac{10-D}{9}$$

In FSRS 5, $D_0(4)$ is the target of mean reversion. Earlier versions had $D_0(3)$ as the target.

$$D'' = w_7 \cdot D_0(4) + (1-w_7) \cdot D'.$$

The other formulas are the same as FSRS-4.5.

FSRS-4.5

Default parameters

```
[0.4872, 1.4003, 3.7145, 13.8206, 5.1618, 1.2298, 0.8975, 0.031, 1.6474, 0.1367,
```

Formula

> The w_i denotes w[i]. This version uses 17 parameters. The memory state is represented by Stability (S) and Difficulty (D).

The formula of forgetting curve is changed in this update.

The retrievability after t days since the last review:

$$R(t,S) = \left(1 + FACTOR \cdot \frac{t}{S}\right)^{DECAY},$$

where $R(t,S) = 0.9$ when $t = S$.

The next interval can be calculated by solving for t in the above equation after putting the request retention in place of R:

$$I(r,S) = \frac{S}{FACTOR} \cdot \left(r^{\frac{1}{DECAY}} - 1 \right),$$

where $I(r,S) = S$ when $r = 0.9$.

In FSRS v4, $DECAY = -1$ and $FACTOR = \frac{1}{9}$

In FSRS-4.5, $DECAY = -0.5$ and $FACTOR = \frac{19}{81}$

The new forgetting curve drops sharply before S and flatly after S.

FSRS v4

Default parameters

```
[0.4, 0.6, 2.4, 5.8, 4.93, 0.94, 0.86, 0.01, 1.49, 0.14, 0.94, 2.18, 0.05, 0.34, 6,
```

Formula

> The w_i denotes w[i]. This version uses 17 parameters. The memory state is represented by Stability (S) and Difficulty (D).

The initial stability after the first rating:

$$S_0(G) = w_{G-1}.$$

For example, $S_0(1) = w_0$ is the initial stability when the first rating is `again`. When the first rating is `easy`, the initial stability is $S_0(4) = w_3$.

The initial difficulty after the first rating:

$$D_0(G) = w_4 - (G-3) \cdot w_5,$$

where the $D_0(3) = w_4$ when the first rating is `good`.

The new difficulty after review:

$$D'(D,G) = w_7 \cdot D_0(3) + (1 - w_7) \cdot (D - w_6 \cdot (G - 3)).$$

It will calculate the new difficulty with $D' = D - w_6 \cdot (G - 3)$, and then apply the mean reversion $w_7 \cdot D_0(3) + (1 - w_7) \cdot D'$ to avoid "ease hell".

The retrievability after t days since the last review:

$$R(t,S) = \left(1 + \frac{t}{9 \cdot S}\right)^{-1},$$

where $R(t,S) = 0.9$ when $t = S$.

The next interval can be calculated by solving for t in the above equation after putting the request retention in place of R:

$$I(r,S) = 9 \cdot S \cdot \left(\frac{1}{r} - 1\right),$$

where $I(r,S) = S$ when $r = 0.9$.

The new stability after a successful review (the user pressed "Hard", "Good" or "Easy"):

$$\begin{aligned} S_r'(D,S,R,G) = \quad & S \\ & \cdot (e^{w_8} \\ & \cdot (11 - D) \\ & \cdot S^{-w_9} \\ & \cdot (e^{w_{10} \cdot (1 - R)} - 1) \\ & \cdot w_{15}(\text{if G} = 2) \\ & \cdot w_{16}(\text{if G} = 4) \\ & + 1). \end{aligned}$$

Let $SInc$ (the increase in stability) denotes $\frac{S_r'(D,S,R,G)}{S}$, which is equivalent to Anki's factor.

1. The larger the value of D, the smaller the $SInc$ value. This means that the increase in memory stability for difficult material is smaller than for easy material.
2. The larger the value of S, the smaller the $SInc$ value. This means that the higher the stability of the memory, the harder it becomes to make the memory even more stable.
3. The smaller the value of R, the larger the $SInc$ value. This means that the spacing effect accumulates over time.
4. The value of $SInc$ is always greater than or equal to 1 if the review was successful.

In FSRS, a delay in reviewing (i.e., overdue reviews) affects the next interval as follows:

As the delay increases, retrievability (R) decreases. If the review was successful, the subsequent stability (S) would be higher, according to point 3 above. However, instead of increasing linearly with the delay like the SM-2/Anki algorithm, the subsequent stability converges to an upper limit, which depends on your FSRS parameters

image

You can modify them in this playground: https://www.geogebra.org/calculator/ahqmqjvx.

The stability after forgetting (i.e., post-lapse stability):

$$S_f'(D,S,R) = \begin{aligned} & w_{11} \\ & \cdot D^{-w_{12}} \\ & \cdot ((S+1)^{w_{13}} - 1) \\ & \cdot e^{w_{14} \cdot (1-R)} \end{aligned}.$$

For example, if $D = 2$ and $R = 0.9$, with default parameters, $S_f'(S = 100) = 2 \cdot 2^{-0.2} \cdot ((100+1)^{0.2} - 1) \cdot e^{1(1-0.9)} \approx 3$ and $S_f'(S = 1) \approx 0.3$.

FSRS v3

Default parameters

```
[0.9605, 1.7234, 4.8527, -1.1917, -1.2956, 0.0573, 1.7352, -0.1673, 1.065, 1.890
```

Formula

> The w_i denotes w[i]. This version uses 13 parameters. The memory state is represented by Stability (S) and Difficulty (D).

The initial stability after the first rating:

$$S_0(G) = w_0 + (G-1) \cdot w_1,$$

where the $S_0(1) = w_0$ is the initial stability when the first rating is `again`. When the first rating is `easy`, the initial stability is $S_0(4) = w_0 + 3 \cdot w_1$.

The initial difficulty after the first rating:

$$D_0(G) = w_2 + (G - 3) \cdot w_3,$$

where the $D_0(3) = w_2$ when the first rating is `good` .

The new difficulty after review:

$$D^{'}(D, G) = w_5 \cdot D_0(3) + (1 - w_5) \cdot (D + w_4 \cdot (G - 3)) .$$

It will calculate the new difficulty with $D^{'} = D + w_4 \cdot (G - 3)$, and then apply the mean reversion $w_5 \cdot D_0(3) + (1 - w_5) \cdot D^{'}$ to avoid "ease hell".

The retrievability of t days since the last review:

$$R(t, S) = 0.9^{\frac{t}{S}},$$

where $R(t, S) = 0.9$ when $t = S$.

The next interval can be calculated by solving for t in the above equation after putting the request retention in place of R.

$$I(r, S) = S \cdot \frac{\ln(r)}{\ln(0.9)},$$

where $I(r, S) = S$ when $r = 0.9$.

Note: the intervals after Hard and Easy ratings are calculated differently. The interval after Easy rating will multiply `easyBonus` . The interval after Hard rating is `lastInterval` multiplied `hardInterval` .

The new stability after recall:

$$S_r^{'}(D, S, R) = S \cdot (e^{w_6} \cdot (11 - D) \cdot S^{w_7} \cdot (e^{w_8 \cdot (1 - R)} - 1) + 1) .$$

Let $SInc$ (the increase in stability) denotes $\frac{S_r^{'}(D, S, R)}{S}$, which is equivalent to Anki's factor.

1. The larger the value of D, the smaller the $SInc$ value. This means that the increase in memory stability for difficult material is smaller than for easy material.
2. The larger the value of S, the smaller the $SInc$ value. This means that the higher the stability of the memory, the harder it becomes to make the memory even more stable.

3. The smaller the value of R, the larger the $SInc$ value. This means that the spacing effect accumulates over time.
4. The value of $SInc$ is always greater than or equal to 1 if the review was successful.

The following 3D visualization could help understand.

image

In FSRS, a delay in reviewing (i.e., overdue reviews) affects the next interval as follows:

As the delay increases, retrievability (R) decreases. If the review was successful, the subsequent stability (S) would be higher, according to point 3 above. However, instead of increasing linearly with the delay like the SM-2/Anki algorithm, the subsequent stability converges to an upper limit, which depends on your FSRS parameters.

The stability after forgetting (i.e., post-lapse stability):

$$S_f'(D,S,R) = w_9 \cdot D^{w_{10}} \cdot S^{w_{11}} \cdot e^{w_{12} \cdot (1-R)}.$$

For example, if $D = 2$ and $R = 0.9$, with default parameters, $S_f'(S = 100) = 2 \cdot 2^{-0.2} \cdot 100^{0.2} \cdot e^{1(1-0.9)} \approx 5$ and $S_f'(S = 1) \approx 2$.

You can play the function in post-lapse stability - GeoGebra.

image

FSRS v2

Default parameters

```
[1, 1, 1, -1, -1, 0.2, 3, -0.8, -0.2, 1.3, 2.6, -0.2, 0.6, 1.5]
```

Formula

> The w_i denotes w[i]. This version uses 14 parameters. The memory state is represented by Stability (S) and Difficulty (D).

The initial stability after the first rating:

$$S_0(G) = w_0 \cdot (w_1 \cdot (G - 1) + 1)$$

The initial difficulty after the first rating:

$$D_0(G) = w_2 \cdot (w_3 \cdot (G - 4) + 1)$$

The difficulty is clamped within the range $[1, 10]$.

The new difficulty after review:

$$D'(D, G) = w_5 \cdot D_0(3) + (1 - w_5) \cdot (D + w_4 \cdot (G - 3))$$

It first computes a temporary new difficulty, then applies mean reversion towards $D_0(3) = w_2 \cdot (-w_3 + 1)$ to prevent "ease hell". The new difficulty is also clamped within $[1, 10]$.

The new stability after a successful review (the user pressed "Hard", "Good" or "Easy"):

$$S_r'(S, D', R) = S \cdot \left(1 + e^{w_6} \cdot (D')^{w_7} \cdot S^{w_8} \cdot (e^{(1-R) \cdot w_9} - 1)\right)$$

The new stability after forgetting (the user pressed "Again"):

$$S_f'(S, D', R) = w_{10} \cdot (D')^{w_{11}} \cdot S^{w_{12}} \cdot (e^{(1-R) \cdot w_{13}} - 1)$$

FSRS v1

Default parameters

```
[2, 5, 3, -0.7, -0.2, 1, -0.3]
```

Formula

> The w_i denotes w[i]. This version uses 7 parameters. The memory state is represented by Stability (S), Difficulty (D), and Lapses (L), the total number of times the card has been forgotten.

The retrievability of t days since the last review:

$$R(t,S) = 0.9^{\frac{t}{S}}$$

The initial state after the first rating:

- Initial Stability: $S_0(G) = w_0 \cdot 0.25 \cdot 2^{G-1}$
- Initial Difficulty: $D_0(G) = w_1 - (G-3)$
- Initial Lapses: $L_0(G) = \max(0, 2-G)$ (This means L_0 is 1 if the first rating is `Again`, and 0 otherwise.)

The new difficulty after review:

$$D'(D,R,G) = \max(0, D + R - 0.25 \cdot 2^{G-1} + 0.1)$$

The new stability after a successful review (the user pressed "Hard", "Good" or "Easy"):

$$S_r'(S,D',R) = S \cdot \left(1 + e^{w_2} \cdot (D' + 0.1)^{w_3} \cdot S^{w_4} \cdot (e^{(1-R) \cdot w_5} - 1)\right)$$

The new stability after forgetting (the user pressed "Again"):

$$S_f'(L) = w_0 \cdot e^{w_6 \cdot L}$$

In this version, the post-lapse stability only depends on the total number of lapses.

Updating lapses after review:

$$L' = L + \max(0, 2-G)$$

The lapse count is incremented by 1 each time the user presses `Again`.